ZINEDDINE REBBOUH

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Summary

• AI Engineer with 2+ years building LLM-powered products and backend services in Python (Django REST Framework, FastAPI), React/TypeScript and PostgreSQL. Shipped Tawfir, a production RAG assistant (Gemini, pgvector) that resolved 68% of support tickets without a human, with 89% Recall@4 on my own evaluation set, deployed through Docker and GitHub Actions CI/CD. I write Pytest and Playwright tests with every feature, work through Git pull requests and code review, and test every line my AI coding assistants generate before it merges. M.Sc. in Artificial Intelligence. Native Arabic speaker, based in Algeria and relocating to Riyadh, KSA.

Skills

• **AI and LLMs:** LLM APIs (Gemini) | RAG | pgvector | Retrieval Evaluation (Recall@4) | Human Escalation Flows | Computer Vision (YOLOv8, 3D CNN) | scikit-learn | XGBoost | AI-Assisted Development
• **Backend and APIs:** Python | FastAPI | Django REST Framework | Node.js/Express | REST API Design | WebSockets | JWT | RBAC | Stripe API | Celery | Django Channels | Async Processing
• **Databases:** PostgreSQL/PostGIS | pgvector | MongoDB | Redis | Schema Design | Indexing | Query Optimization
• **Testing and DevOps:** Git/GitHub (Pull Requests, Code Review) | GitHub Actions CI/CD | Docker | Linux | Pytest | Playwright | Vercel | Render | Agile/Scrum
• **Frontend and Fundamentals:** TypeScript/JavaScript | React | Next.js | Redux Toolkit | SQL | OOP | Data Structures and Algorithms | Distributed Systems
• **Languages:** Arabic (native) | English (professional) | French (intermediate)

Experience

Freelance AI and Backend Developer — **Self-employed** — *Remote* — **Mar 2024 - Present**

• Build and ship backend services, LLM features and web apps for independent clients (Django REST Framework, FastAPI, Node.js/Express, React), from requirements to deployment and handover docs.
• Built and deployed Tawfir (RAG assistant, Celery recommender, REST/PostGIS backend) over 6 sprints on Vercel and Render, with Docker and GitHub Actions CI/CD.
• Write Pytest and Playwright tests plus CI smoke checks for every engagement, run code reviews, and tune performance (caching, lazy loading, CDN).

Machine Learning Engineer Intern — **Elevvo Pathways** — *Remote* — **Jul 2025 - Aug 2025**

• Built 6 ML pipelines in 2 months (forecasting, recommendation, computer vision), including an XGBoost sales forecaster tuned with 5-fold GridSearchCV, documented in reproducible code and reports.

React.js Developer Intern — **Codsoft** — *Remote* — **Mar 2024 - Apr 2024**

• Built 3 React.js apps in a 4-week sprint, with a shared component library and custom hooks.

Projects

• **Tawfir (AI):** Flutter app and Django REST/PostGIS backend that reads freshness from a photo and matches surplus stock with nearby buyers. RAG assistant (Gemini 2.5 Flash, pgvector) at 89% Recall@4, resolving 68% of tickets without a human (30 evaluated conversations); Celery recommender; Redis responses under 80 ms.
• **Sign Language Recognition (AI):** Real-time sign-to-text with a 3D CNN (14.3M parameters, 90.71% validation accuracy), served with FastAPI and Docker; GitHub Actions runs Pytest and container tests per PR.
• **Football Analytics (AI):** Player speed and positioning from match footage (YOLOv8, ByteTrack), 95%+ ID consistency at 50 to 60 FPS.
• **Domainly AI:** Embeddable AI chatbot for small businesses that qualifies leads, books appointments, takes payments, and hands the conversation to a human when needed.
• **WANLP (AI):** Real-time Arabic sports trend detection: 1,000+ social media events per minute, sub-second updates over Django Channels WebSockets to a Next.js/TypeScript dashboard (Celery, MongoDB, Redis).
• **HotelPro:** Booking and hotel management app (admin, staff, guest dashboards), Stripe payments, 20+ REST endpoints secured with JWT/RBAC, Playwright E2E tests.
• **Multi-Vendor E-Commerce:** MERN marketplace with vendor storefronts, one checkout across shops, JWT auth and Stripe payments.
• **Habitat:** Real-estate rental web app with a marketing landing page and a tenant dashboard to browse listings, view property details, save favorites, and track applications, residence, billing and payment methods.

Education

University of Constantine 2 — *CONSTANTINE, ALGERIA* — **2021 - 2026**

• **M.Sc. Artificial Intelligence and Data Science** (2024 - 2026): Thesis on Tawfir, an AI platform in production (Django REST, PostgreSQL, RAG). Grade: 19/20.
• **B.Sc. Computer Science and IT** (2021 - 2024): Final-year marketplace project ranked 1st in the class (17/20), defended in English. Coursework: Distributed Systems, Databases, Networks, Network Security.